

Technical Appendix: Sparse LD and CSR

Peter Sørensen

Table of contents

This appendix defines the sparse linkage disequilibrium (LD) representation used by the current `sblr` BED-to-CSR workflow. It describes the statistical matrix, compressed sparse row (CSR) structure, disk-backed construction, indexing conventions, alignment requirements, and downstream conversion used by `stblr_csr()` and related wrappers.

The stable user-facing contract is deliberately narrower than the current binary-file implementation:

1. create sparse LD with `make_sparseLD()`, which records metadata in `Glist$sparseLD`;
2. inspect the resulting prefix with `sparseLD_read_CSR()` when needed; and
3. pass the updated `Glist` to `stblr_csr()` or `stblr_csr_annot()`.

`sparseLD_stream_CSR()` is the lower-level writer used by the preparation helper. Its `ld_prefix` remains the disk-backed artifact identifier, but the current high-level workflow carries that prefix through `Glist$sparseLD`.

The present file suffixes, integer widths, floating-point widths, metadata keys, and upper-triangle storage convention are documented here to explain behavior and support validation. They should not be treated as a guaranteed manual file-format API. Use package readers, writers, and samplers rather than editing or independently constructing these files.

1 From BED Genotypes to Sufficient Statistics and Sparse LD

For genotype matrix X , phenotype vector y , and marker effects b , a summary-statistics marker sampler needs quantities including

$$wy = X^\top y, \quad ww_i = x_i^\top x_i, \quad yy = y^\top y,$$

and enough off-diagonal marker information to update

$$r = X^\top (y - Xb) = wy - X^\top Xb.$$

In the BED-derived workflow, these inputs are prepared in two coordinated branches: